

EIR Foundation Model (EIR-FM)

Integrative Post-GWAS Methods

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**For this session, you will need the
'eir-fm-combined.ipynb' notebook.**

Overview

1. Polygenic scores.
2. Foundation models. One model, many traits. What that gives you and what it does not.
3. Prediction on PennCATH; running eirautogppredict.
4. Evaluation. Correlation, incremental R^2 , ROC and PR-AUC, trait clustering, PCA and UMAP, downstream CAD prediction with feature selection and bootstrap CIs, and SHAP.

Part 1

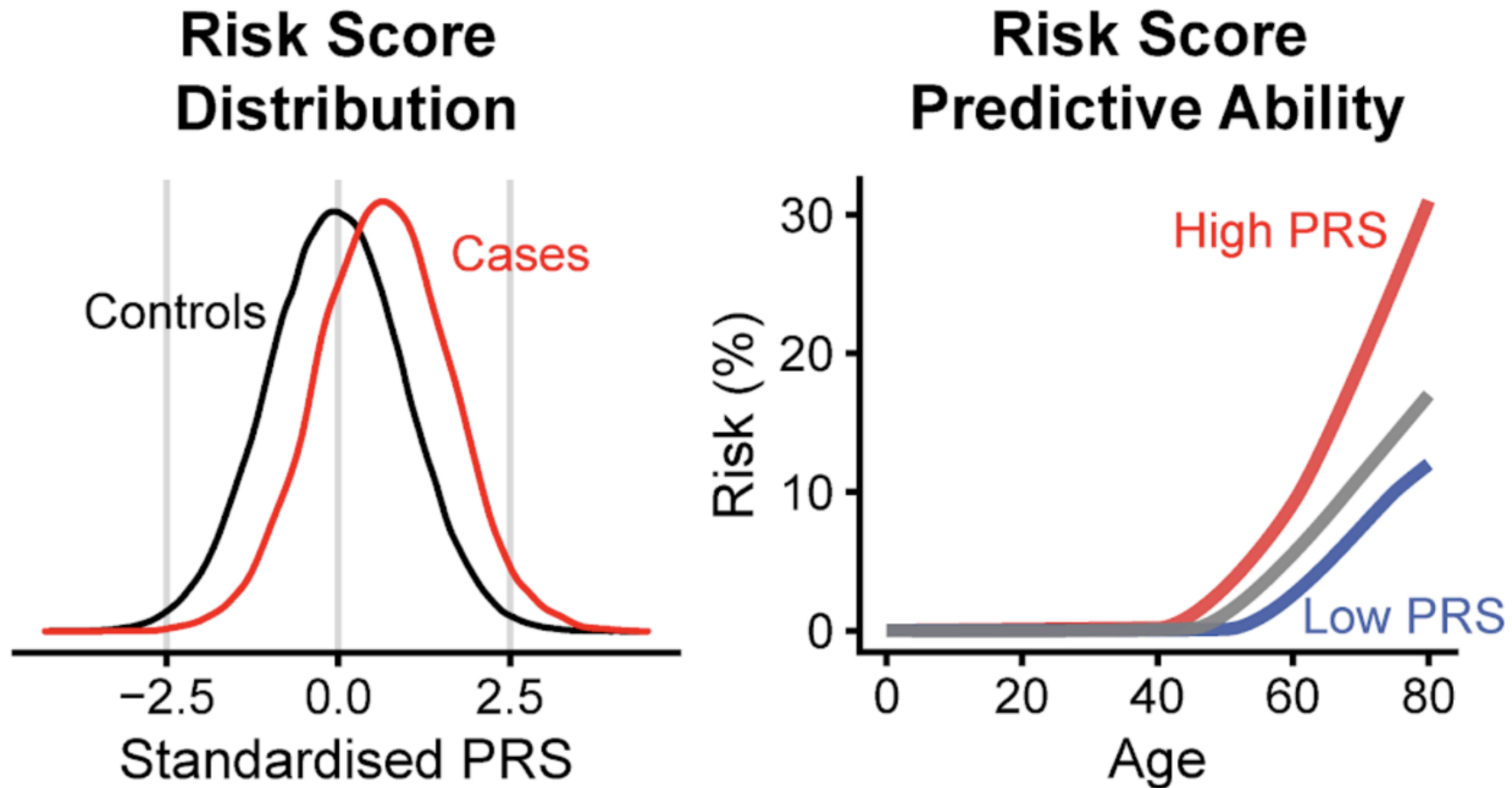
Polygenic scores

Assumed familiar, so we move through this quickly

What a polygenic score is ...

- One number per person, summarizing many variants for one trait
- GWAS gives an effect size β per variant. Dosage is the number of effect alleles carried (0, 1 or 2). The score is a weighted sum:
 - $\text{PGS} = \sum \beta_i \times \text{dosage}_i$
- Classical PGS methods differ only in which weights they use

Many small effects, one score



Why the naïve sum fails

- Linkage disequilibrium: nearby variants tag the same signal
 - summing them double-counts the effect
- Most effect estimates are small and noisy
 - including them unfiltered usually hurts prediction
- C+T, LDpred2 and PRS-CS exist to handle exactly these two
 - LDpred2 tutorial: <https://privefl.github.io/bigsnpr/articles/LDpred2.html>

Part 2

From PGS to foundation model

Two ways to get genetic predictions

Classical PGS: one trait at a time

- One GWAS per trait
- Derive a scoring rule
- Apply it, get one score
- Want LDL too? Start over

Foundation model: all traits at once

- One network, many traits
- Learns a shared genome view
- Hundreds of scores at once
- New trait costs nothing

What a genomic foundation model can and cannot do

- **PGS-style use.** use the output directly as a polygenic score and evaluate it as usual. We do this for HDL, LDL, TG and CAD today.
- **Phenome-wide prediction.** no extra development cost for trait number 300.
- **Multi-trait profiling.** Instead of asking for one person's risk of disease X, you can ask what their whole genetic profile looks like. That enables clustering, outlier detection and subtyping.
- **(Possibly) non-additive effects.** Neural networks can in principle capture dominance and epistasis that additive PGS cannot, though this is debated.
- **Not a replacement.** For a single additive trait with a well-powered GWAS, a well-tuned LDpred2 or PRS-CS often matches or beats a multi-task model. These are complementary tools.

Today's setup

- We train nothing today. Training needs biobank-scale data and a GPU. We download a pre-trained model, apply it to PennCATH, and analyse the output.
- 1,401 PennCATH individuals: measured CAD status, age, sex, TG, HDL, LDL
- ~500K UK Biobank training set
- 315 predicted phenotypes: 237 continuous and 78 binary, per individual
- **Transfer what was learned in the UKB into PennCATH data.**

Part 3

Running the EIR-FM model

Data setup: two downloads

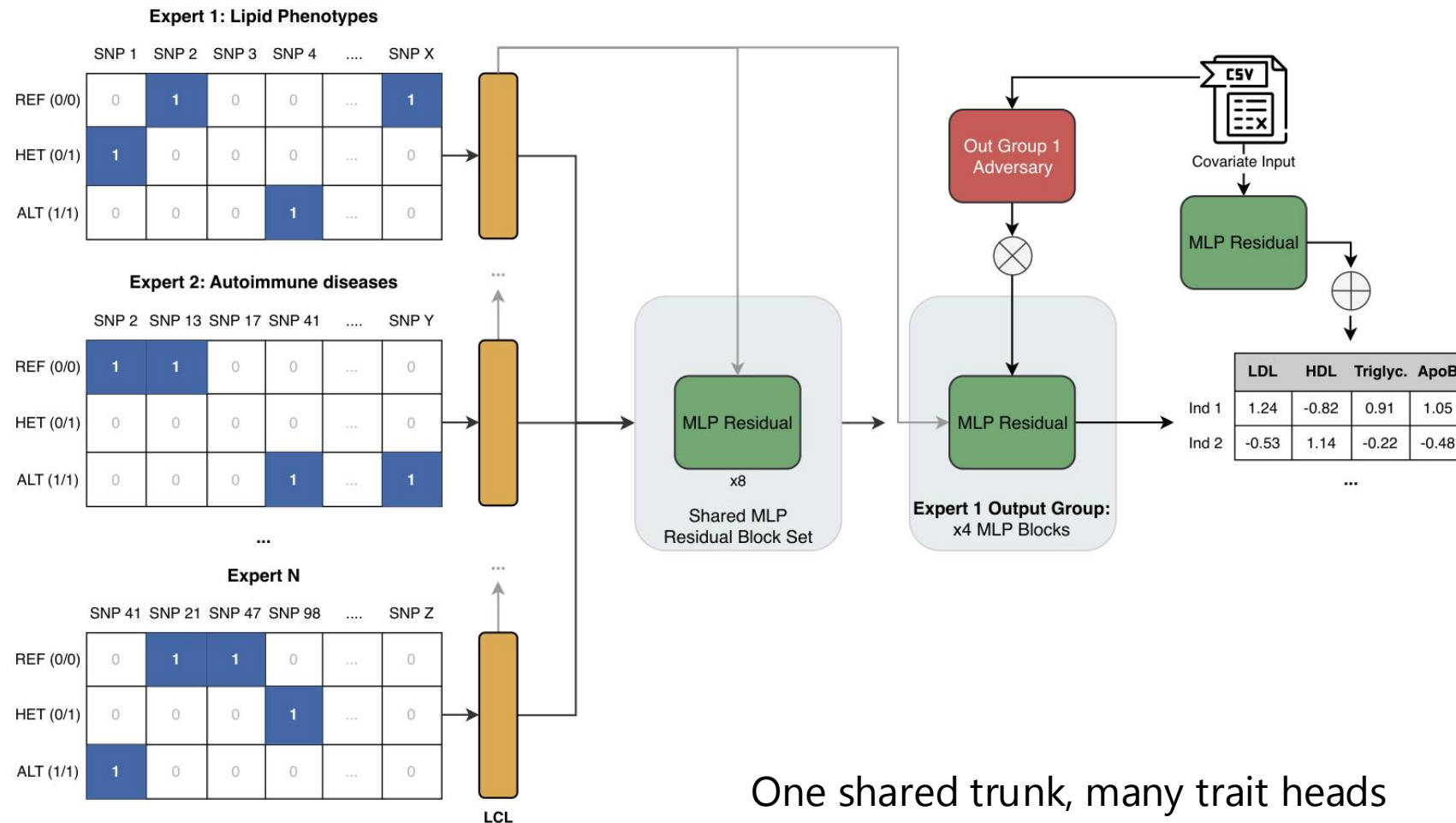
- The file `ext_d_inf_moe_general_16L-FUS_multitask.zip` holds the trained network weights.

```
model_zip_id = "1hRnULP7o8nOdWUzp4uJQcqnKZakuFJ3W"  
gdown.download(id=model_zip_id, output="../reference_data/models/")
```

```
penncath_hg38_id = "1d1phne1USPa7gkj7LOneyFH9zcGexvHs"  
gdown.download(id=penncath_hg38_id, output="../reference_data/")
```

```
!unzip -o ../reference_data/penncat_hg38.zip -d ../reference_data
```

The model architecture



Running the model

```
# Pin the EIR version the model was trained with
```

```
import importlib.metadata as m
if not m.version("eir-dl").startswith("0.23"):
    !uv pip install "git+https://github.com/arnor-sigurdsson/EIR.git@b959bfa"
```

- One command produces all 315 predictions
 - The command may sit silent for a few seconds while the deep-learning libraries load. That is normal.

```
!eirautogppredict \  
  --genotype_data_path ../reference_data/pennath_hg38 \  
  --packed_experiment_path ../reference_data/models/ext_d_inf_moe_general_16L-FUS_multitask.zip \  
  --output_folder ../output
```

SNP harmonization

- The model expects the variants it saw during training
- For PennCATH:
 - 861,221 SNPs available in the cohort
 - 94,528 match what the UKB model was trained on, **about 11%**

SNP Overlap Analysis Report

Metric	Value
COVERAGE METRICS	
Original Training SNPs Recovered (Main)	39.5%
New Cohort SNPs Usable (of total)	11.0%*
MATCHING DETAILS	
Total Matching SNPs	94,528
- Exact Matches (A/G -> A/G)	31,513 (33.3%)
- Allele Reversals (A/G -> G/A)	15,743 (16.7%)
- Strand Flips (A/G -> T/C)	31,404 (33.2%)
- Strand Flip + Reversal (A/G -> C/T)	15,868 (16.8%)
DATASET INFO	
Original Training SNPs	239,463
New Cohort SNPs	861,221
Positional Overlaps (before filtering)	94,541
EXCLUDED SNPS	
Ambiguous/Palindromic SNPs (A/T, C/G)	1
Allele Mismatches (different variants)	12
GAP ANALYSIS	
Missing Original SNPs	144,935 (60.5%)
Unused New Cohort SNPs	766,693
QUALITY	
Match Type Quality	Low (<75% SNPs recovered)

Part 4

Analysis

315 predictions per person, and what we can conclude from them

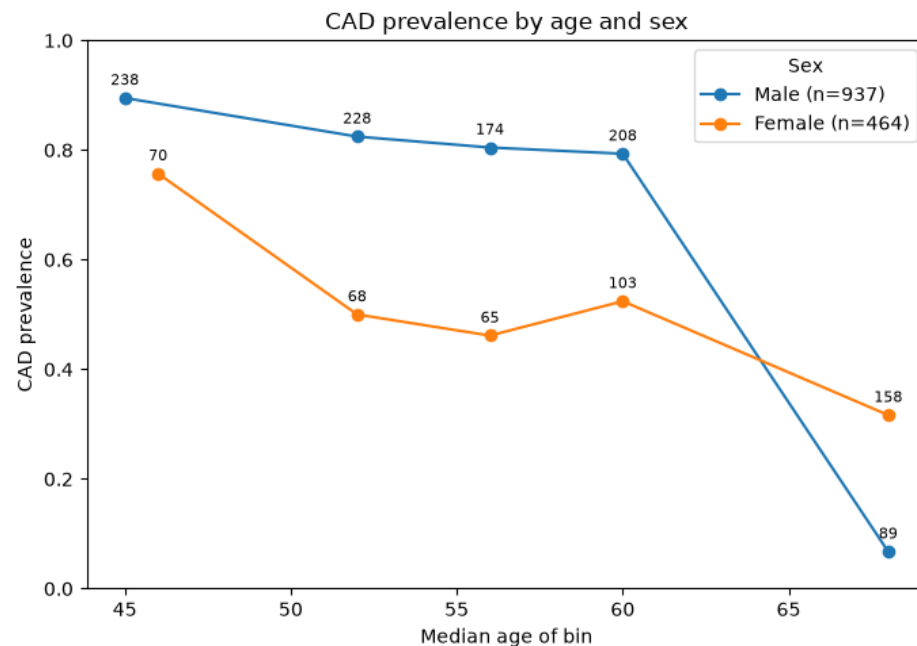
What the model writes out

- One .csv per predicted phenotype, in ../output/results/.
- Each file has an '... Ensemble' column: the average across training folds. Averaging folds also lets you estimate uncertainty on a single prediction.
 - Here there is a single fold, so ... Ensemble equals Fold 0.
- Binary traits instead expose '... Ensemble Prob 1', the predicted probability of the positive class. That is how the loader distinguishes the two kinds.
- Gathering the files gives one matrix:
 - 1,401 individuals × 315 predicted phenotypes

Look at the labels before the model

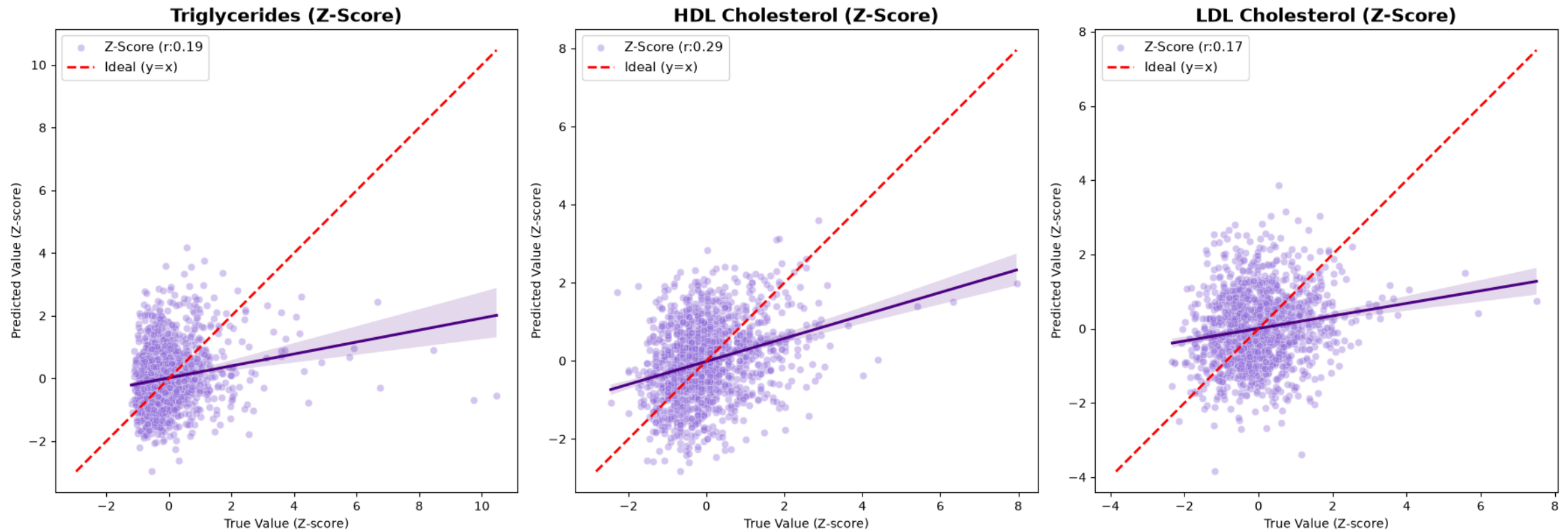
- PennCATH measured phenotypes: CAD status, age, sex, triglycerides, HDL, LDL.
- `labels["CAD"].value_counts()` gives 1: 933 / 0: 468
 - 67% cases. This is not a population sample. It is a case-enriched angiography cohort.
- A model that predicts CAD for everyone scores 67% accuracy while learning nothing. Accuracy is not usable here.

CAD prevalence goes down with age



- The pattern is the opposite of what population epidemiology would give you.
- Cases were recruited young and controls old, to enrich for genetic signal in the cases.
- This returns in the SHAP analysis at the end, where **young** age pushes predicted risk up.

Predicted against measured lipids



Pearson correlation coefficient; Z-scored predictions against z-scored measurements for TG, HDL and LDL.

That looks like weak prediction. Is it?

- These correlations are relatively weak. Reasons that is expected:
 - **This is genetics only.** No age, sex, lifestyle or medication.
 - **Cross-cohort transfer.** The model was trained on UKB and only 11% of its input SNPs are present here.
- So the question is not whether r is large. It is whether this adds information we did not already have.

Incremental R^2 (ΔR^2): how much do predictions improve over a baseline?

- Worry: the model may just have learned age and sex
 - Ex. It will look like it predicts triglycerides from genetics while actually recovering the male-female difference.
- Fit two nested models and compare
 - baseline: $\text{trait} \sim \text{age} + \text{sex}$
 - full: $\text{trait} \sim \text{age} + \text{sex} + \text{prediction}$
- ΔR^2 ; The difference is what the prediction adds on its own
- Put whatever you suspect of confounding into the baseline: ancestry PCs, genotyping batch, recruitment center.

Incremental R^2 in code

```
for trait, full_name in continuous_traits.items():  
    y_col, prs_col = f"{trait}_true", f"{trait}_pred"  
  
    # Base model: the confounders we are worried about  
    model_base = smf.ols(f"{y_col} ~ age + sex", data=df).fit()  
    r2_base = model_base.rsquared * 100  
  
    # Full model: the same, plus the prediction under test  
    model_full = smf.ols(f"{y_col} ~ age + sex + {prs_col}", data=df).fit()  
    r2_full = model_full.rsquared * 100  
  
    delta = r2_full - r2_base           # the number we report
```

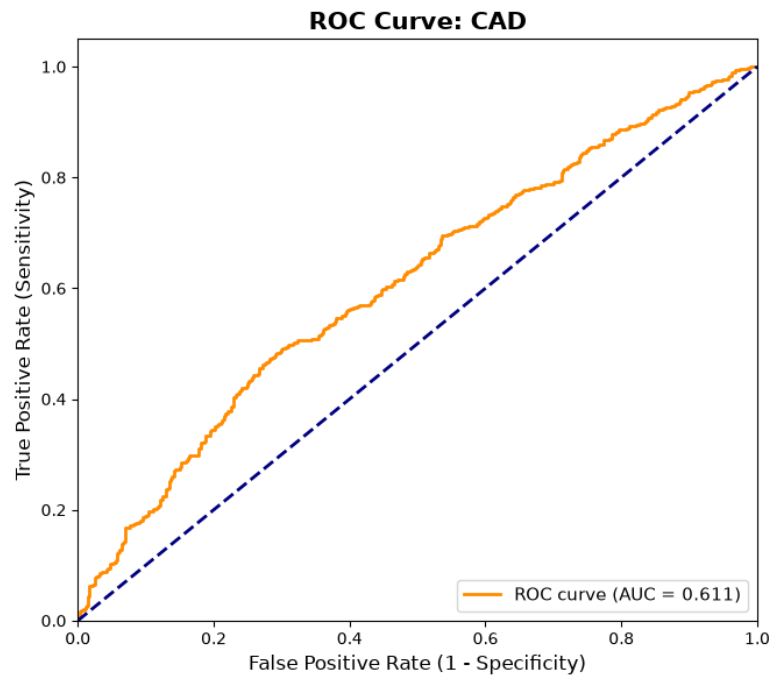
How much do the predictions add?

Trait	Age + sex	Plus prediction	Gain
Triglycerides	2.91%	5.74%	+2.84%
HDL cholesterol	15.60%	22.24%	+6.64%
LDL cholesterol	1.80%	4.70%	+2.89%

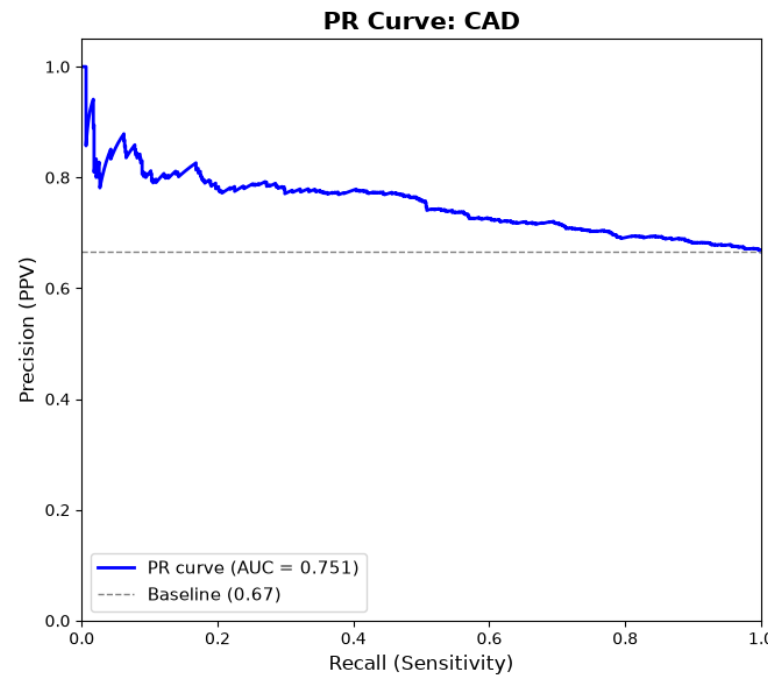
Binary traits need different metrics: why not accuracy?

- We evaluate CAD using the model's I25 output (ICD-10: chronic ischaemic heart disease).
- Accuracy and F1-score need a hard threshold.
- ROC-AUC: do positives score above negatives?
- PR-AUC: of the top-scoring people, how many are real cases?

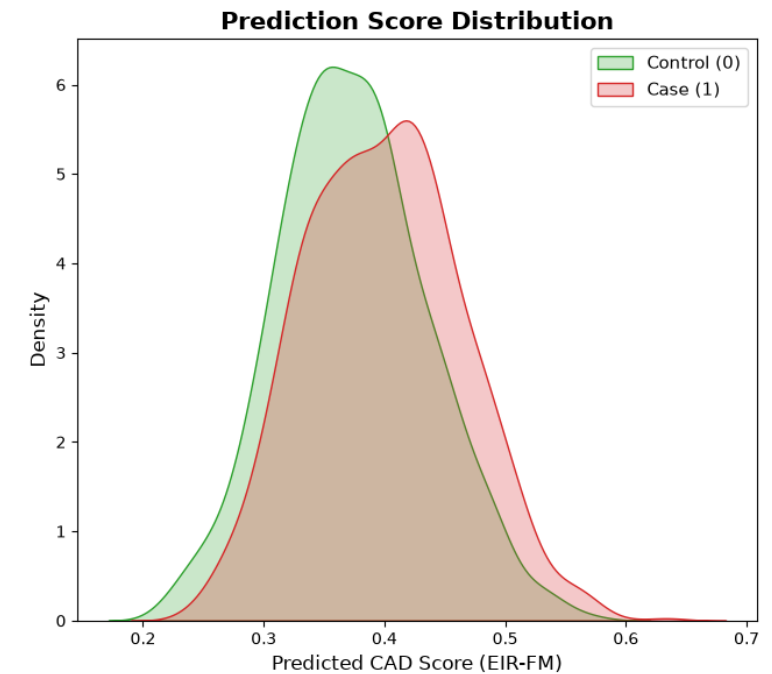
CAD: ROC and precision-recall



ROC-AUC measures how well a model distinguishes between positive and negative cases.

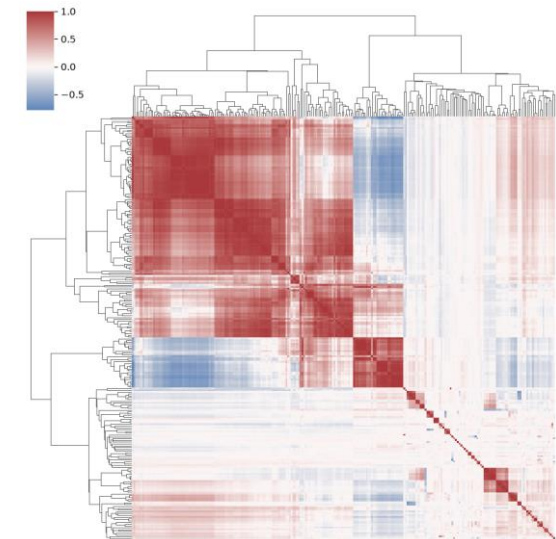


PR-AUC measures how well a model identifies the positive cases while keeping false positives low.



Correlation structure across predicted traits

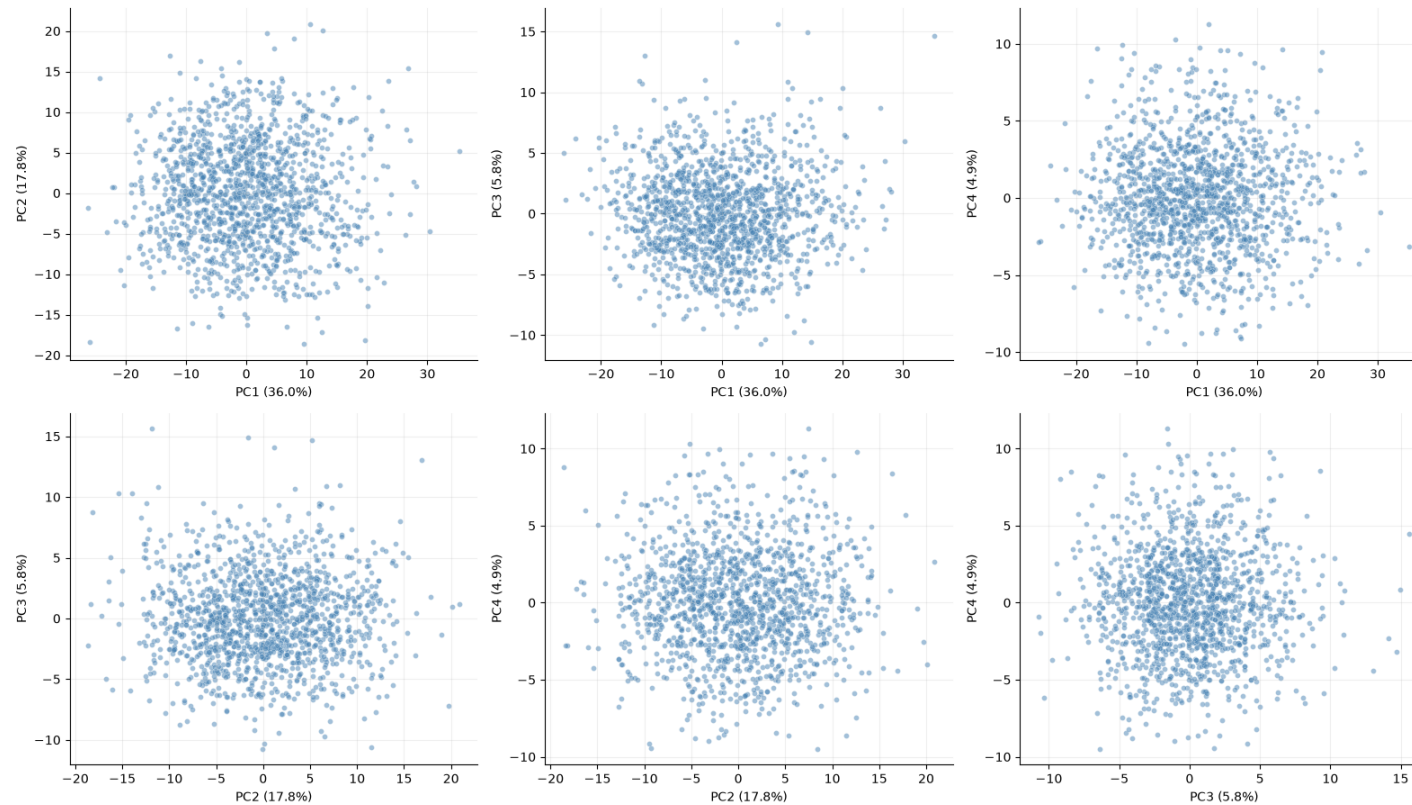
- Sanity check
 - Identifying redundancy
 - Surfacing unexpected structure
-
- Spearman correlation (rank-based, robust to non-linear monotonic relationships and to outliers)



Trait A	Trait B	Spearman correlation
Weight (anthropometric)	Weight (impedance)	0.997
BMI (anthropometric)	BMI (impedance)	0.996
Cholesterol in large HDL	Cholesteryl esters, large HDL	0.997
VLDL particle diameter	Cholesterol in large HDL	-0.777

PCA over PennCATH individuals

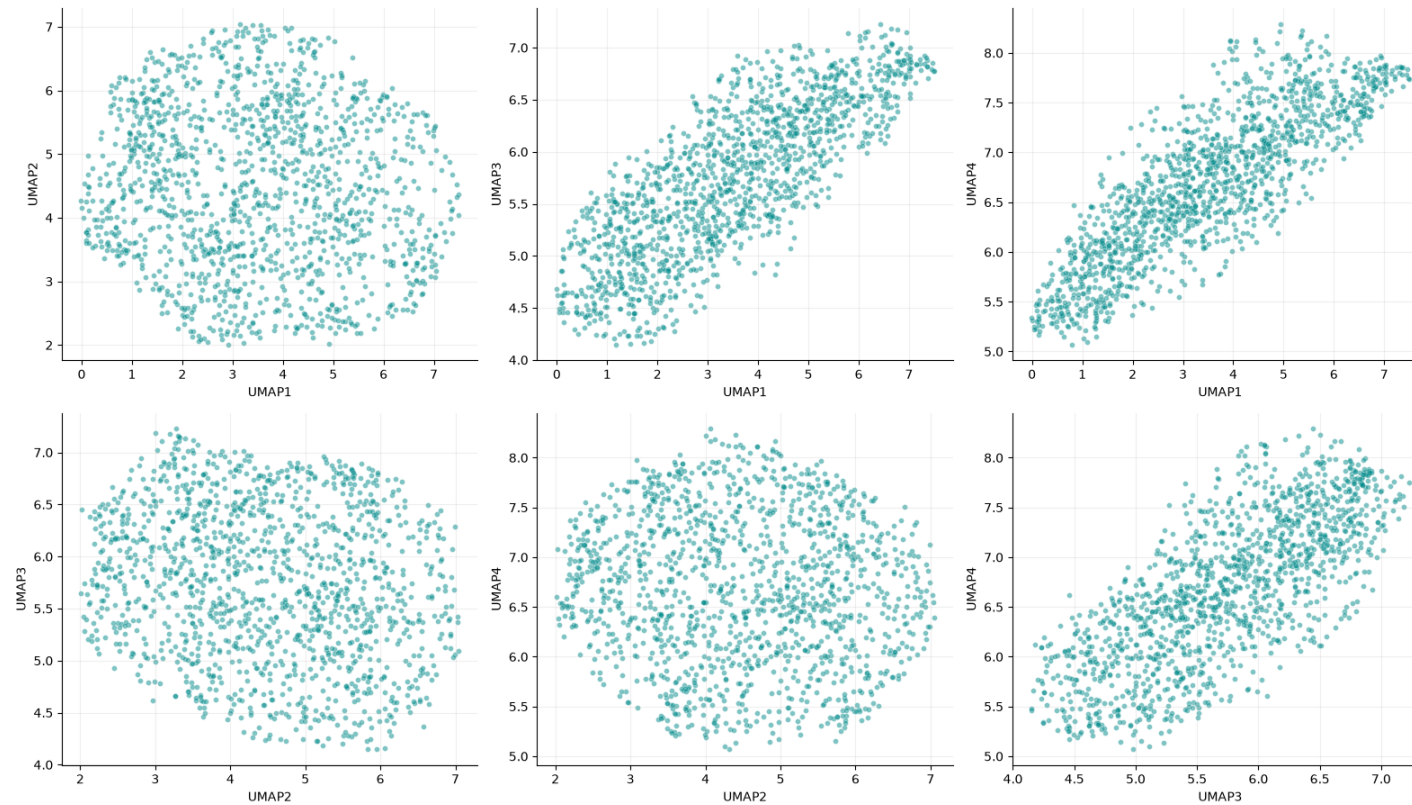
PCA of 237 Continuous-Trait Predictions (n=1401)



All pairwise combinations of PC1 to PC4, each point one individual

UMAP over the same individuals

UMAP Projection of 237 Continuous-Trait Predictions (n=1401)

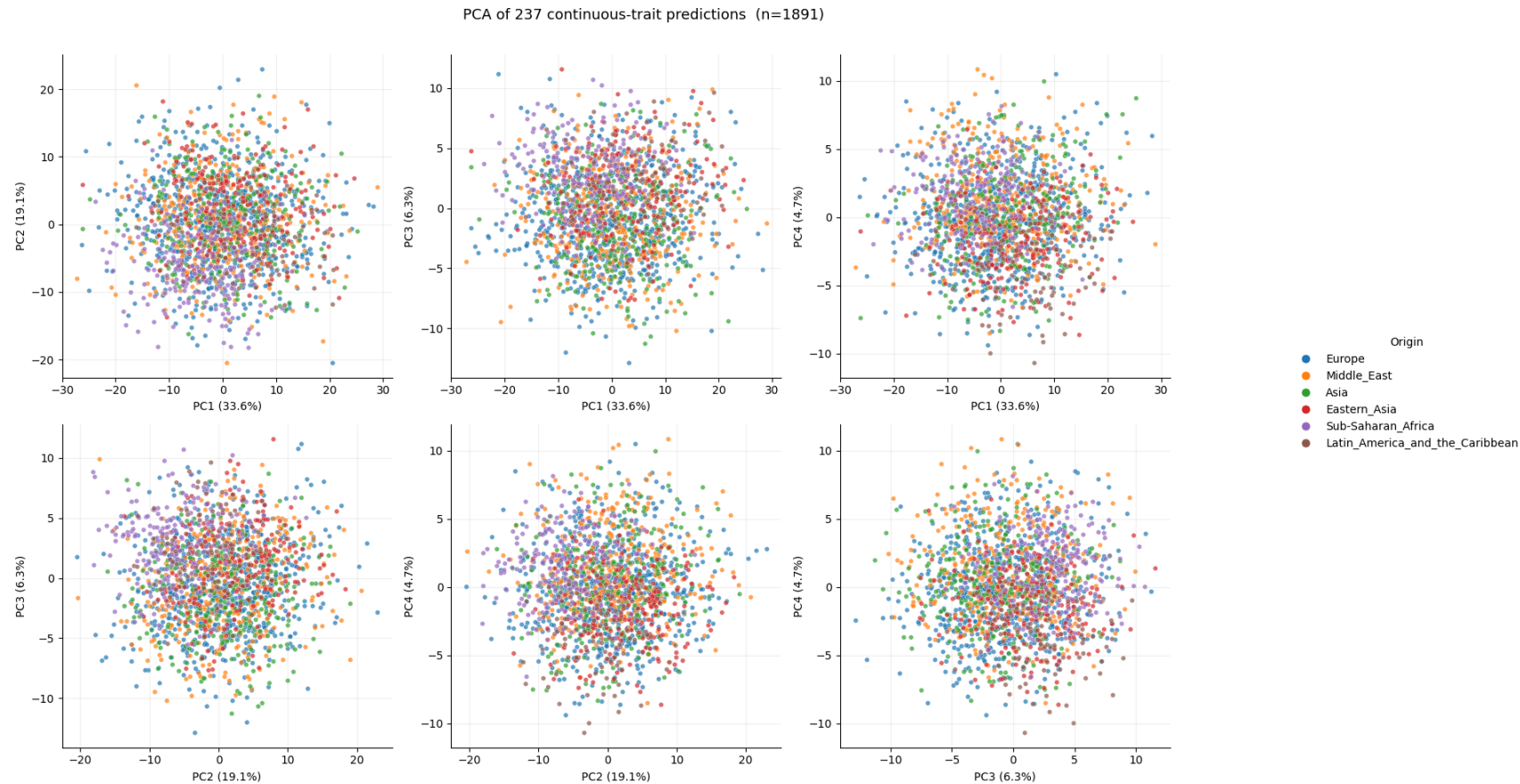


UMAP components 1 to 4, a non-linear view of the same 237-dimensional space

Color the projections before you read them

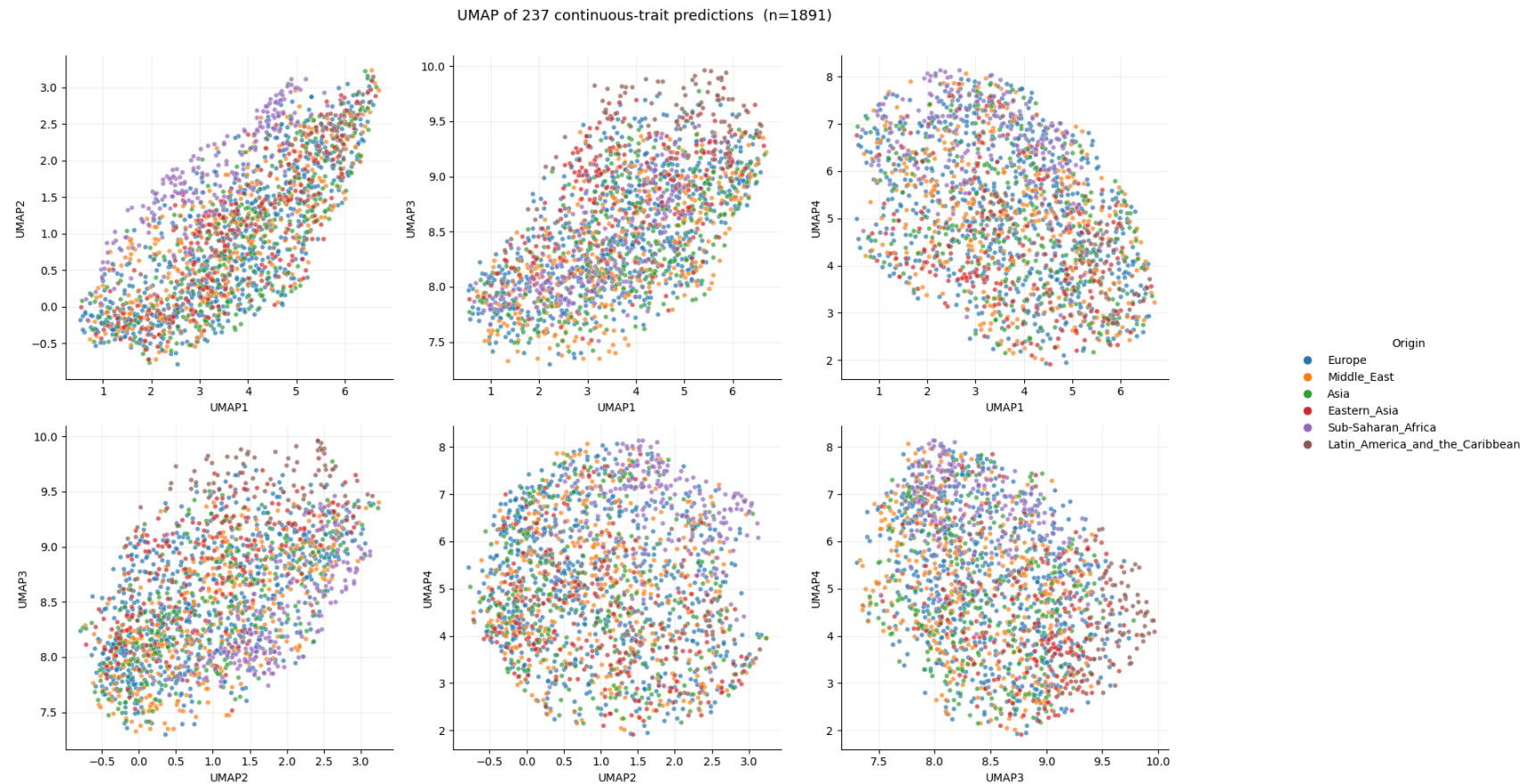
- ***color_by = None*** near the top, with the options written out as comments right above it
 - Measured, from labels: sex (1/2), age, CAD (0/1), hdl, tg, ldl
 - Predicted, from preds: any of the 315 columns, e.g. preds["I25"] or preds["Standing_height"]
 - Set it, re-run the cell.
-
- PennCATH has no ancestry column, which is why the next two slides go to the Human Origins data instead.

Reference: PCA on the Human Origins dataset



The same model's predictions on a dataset spanning many continental backgrounds. If ancestry drove the predictions, populations would separate along the leading PCs

Reference: UMAP on the Human Origins dataset

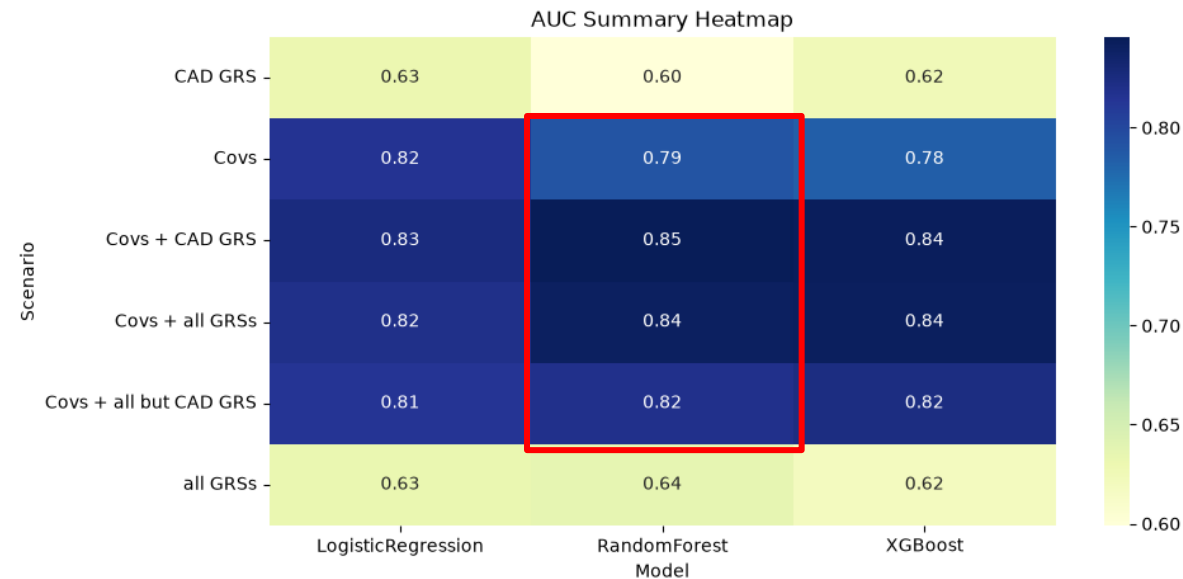


UMAP picks up faint structure PCA missed, but populations still overlap heavily

Can the whole panel improve CAD prediction?

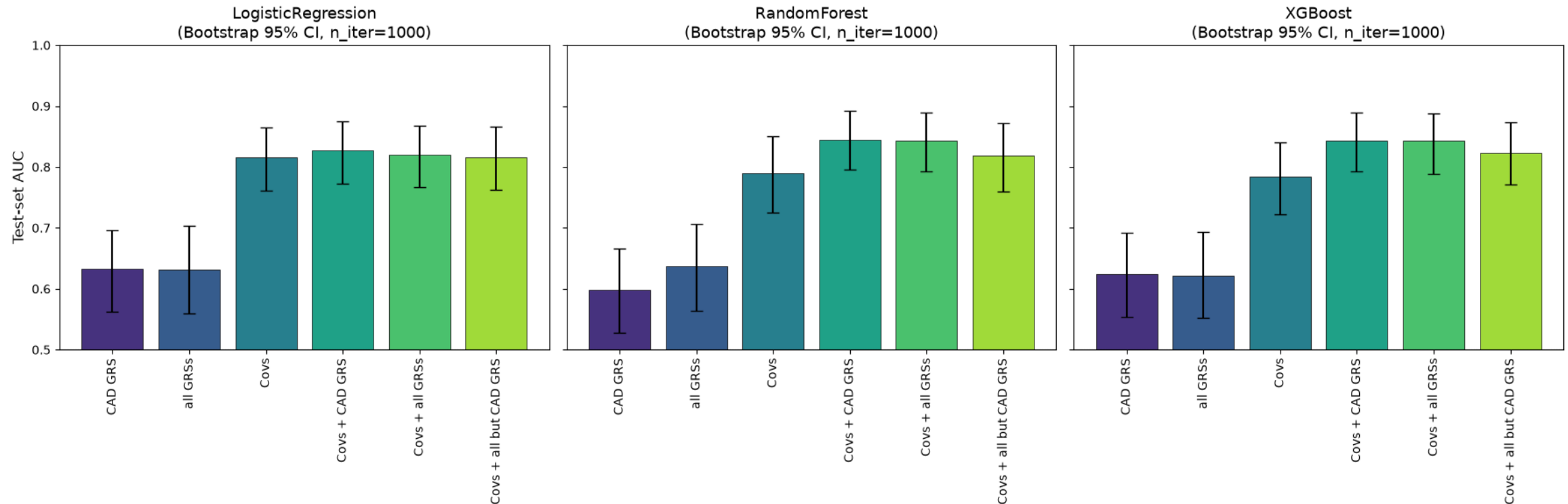
- 315 features and 1,120 training samples will overfit
- So use **Lasso (L1-penalised logistic regression)** as a selector: 315 down to 7 features
- Then three ML classifiers across six feature sets
- Analysis
 - Quantify the uncertainty: Bootstrap the test set before believing any ranking.
 - Examine the winner: ROC, precision-recall and confusion matrix on the held-out test set.
 - Interpret with SHAP: Which features drive the predictions.

Scenario comparison: test-set AUC



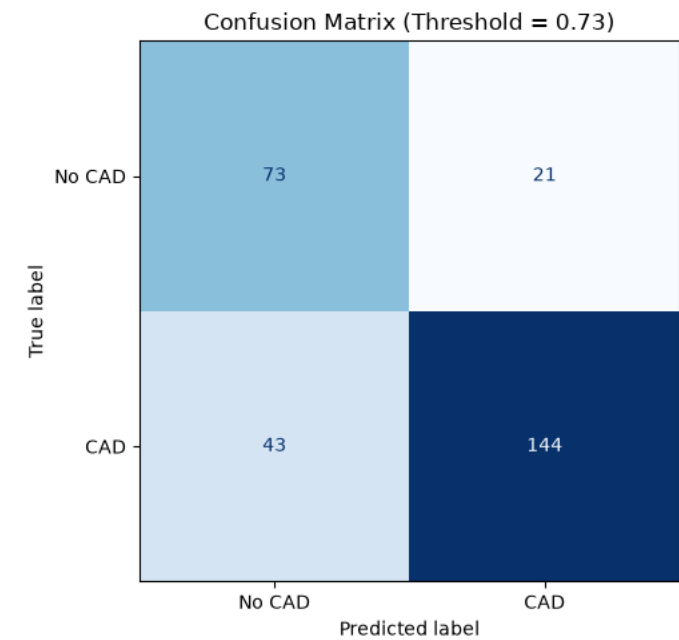
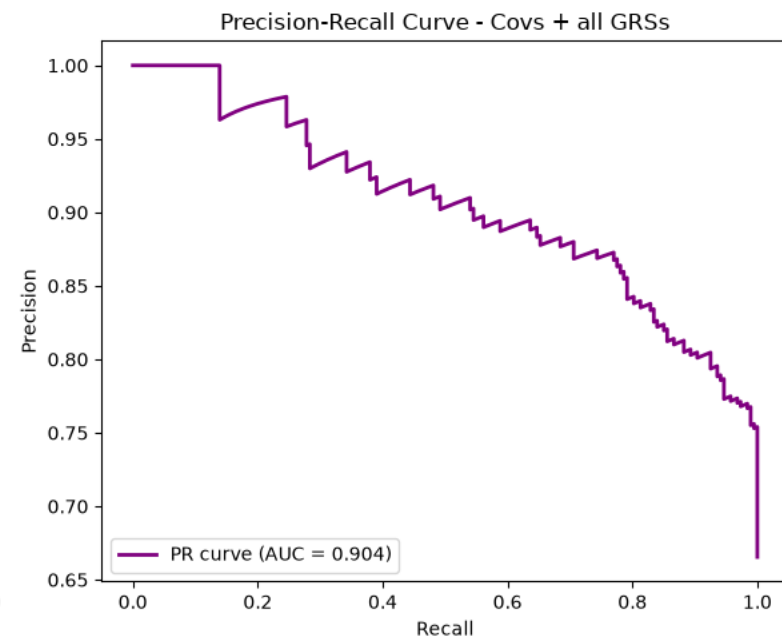
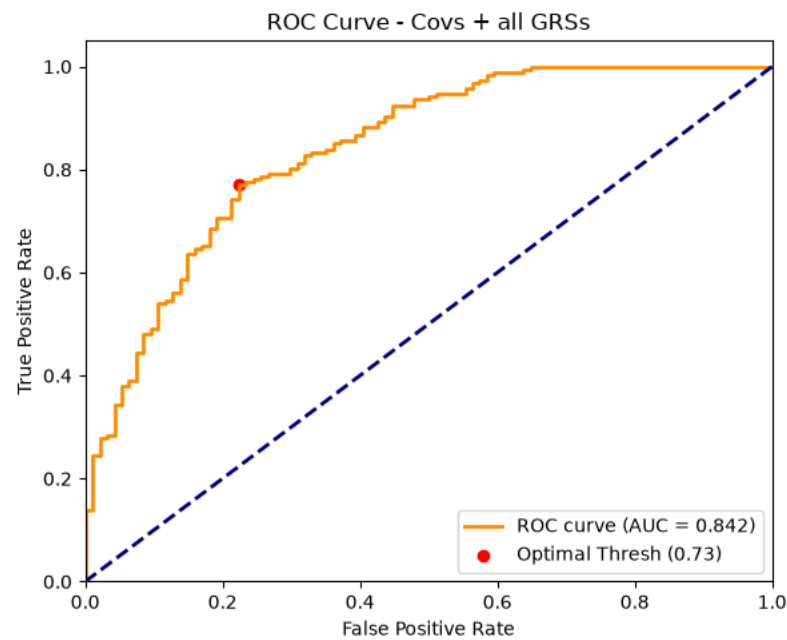
Scenario	Features	What it tests
CAD GRS	1	the single CAD score alone, a classical PRS-only evaluation
all GRSs	7	the selected genetic panel with no demographics
Covs	2	age and sex only, the baseline to beat
Covs + CAD GRS	3	covariates plus one score
Covs + all GRSs	9	everything; does the wider panel add over one score?
Covs + all but CAD GRS	8	can the other traits substitute for the CAD score?

Bootstrap 95% confidence intervals on test AUC



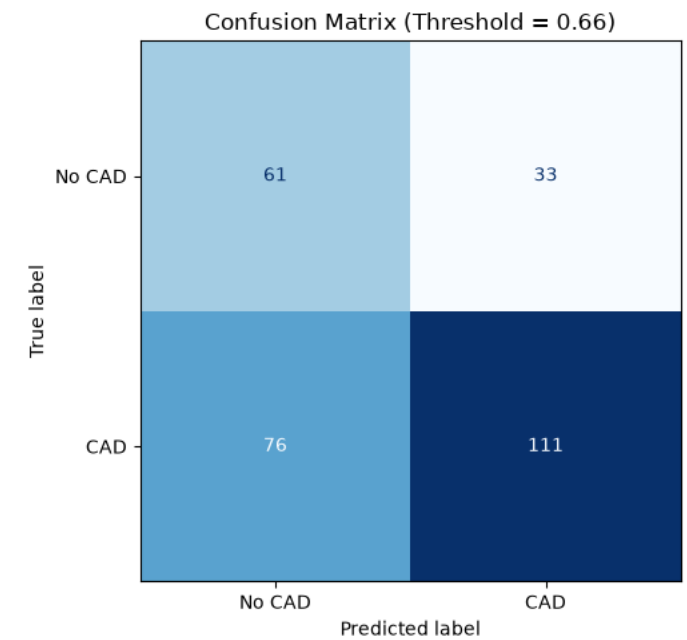
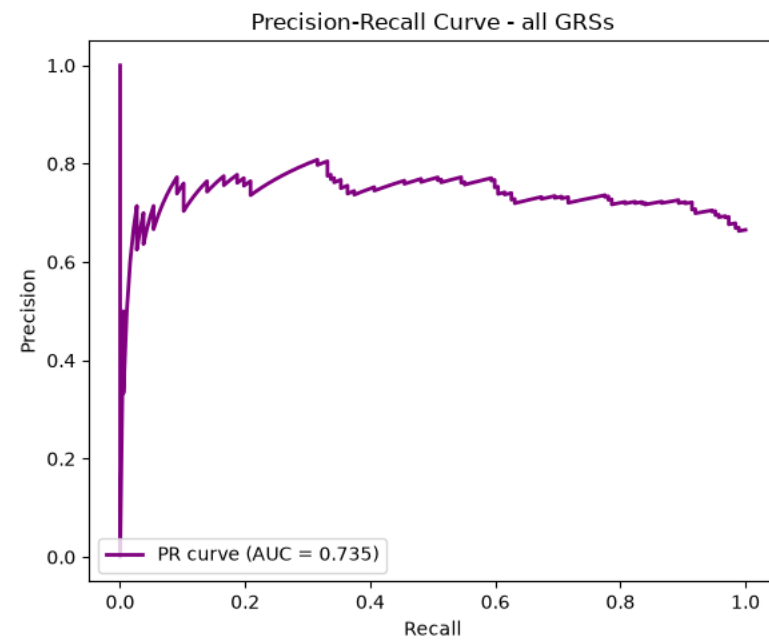
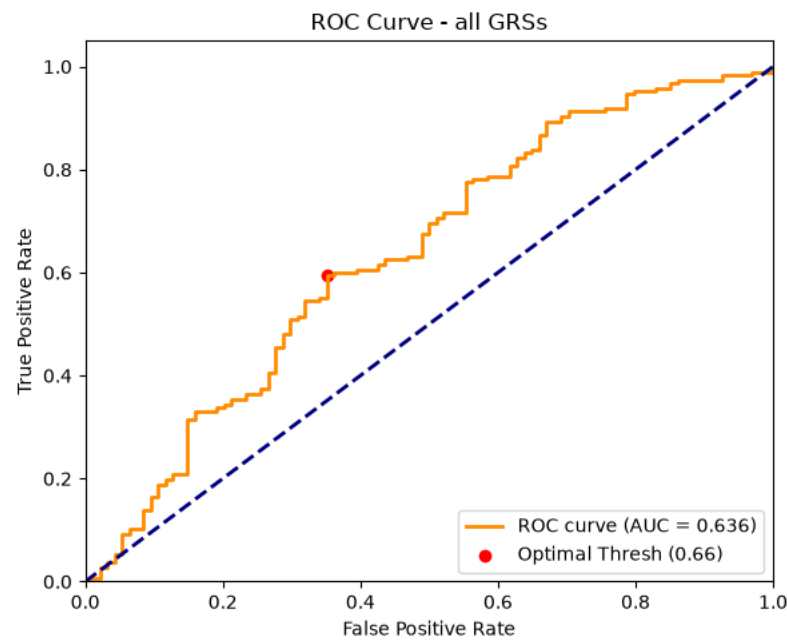
Model	AUC	Pseudo-R ²
Age + sex	0.7779	20.51%
Age + sex + CAD score	0.7881	21.96%
Gain	+0.0102	+1.45%

Best model: ROC, precision-recall, confusion matrix



Covs + all GRSs on the held-out test set. AUC 0.842, PR-AUC 0.904

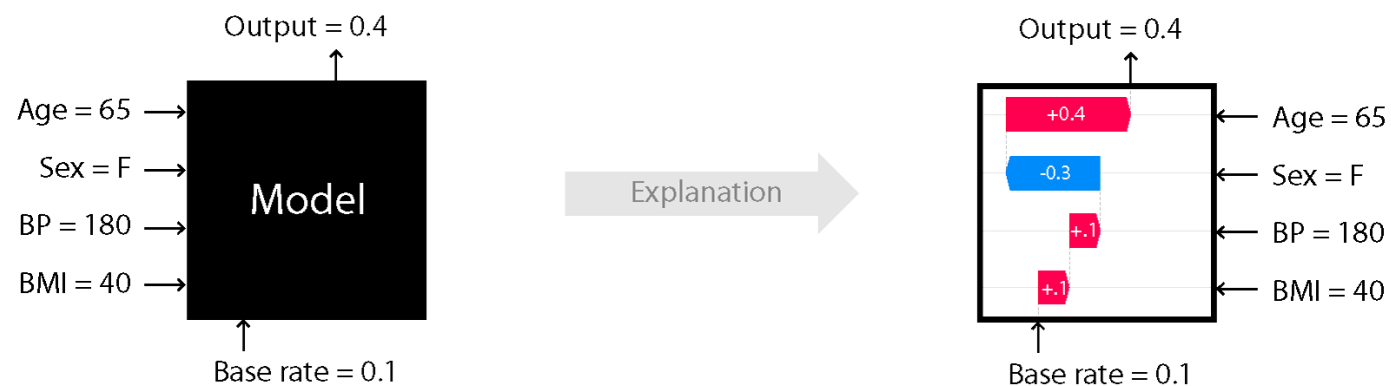
The same model with age and sex removed



all GRSs, no covariates: AUC 0.636, PR-AUC 0.735

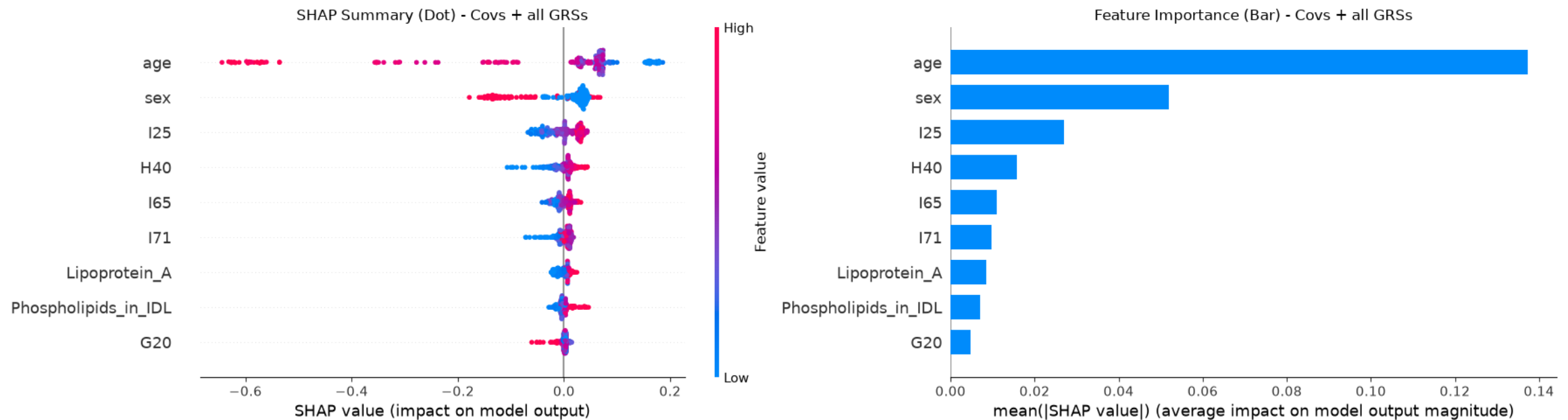
SHAP analysis

- SHAP (SHapley Additive exPlanations) attributes each prediction to its features
 - how much did feature X move this individual's risk away from the average?



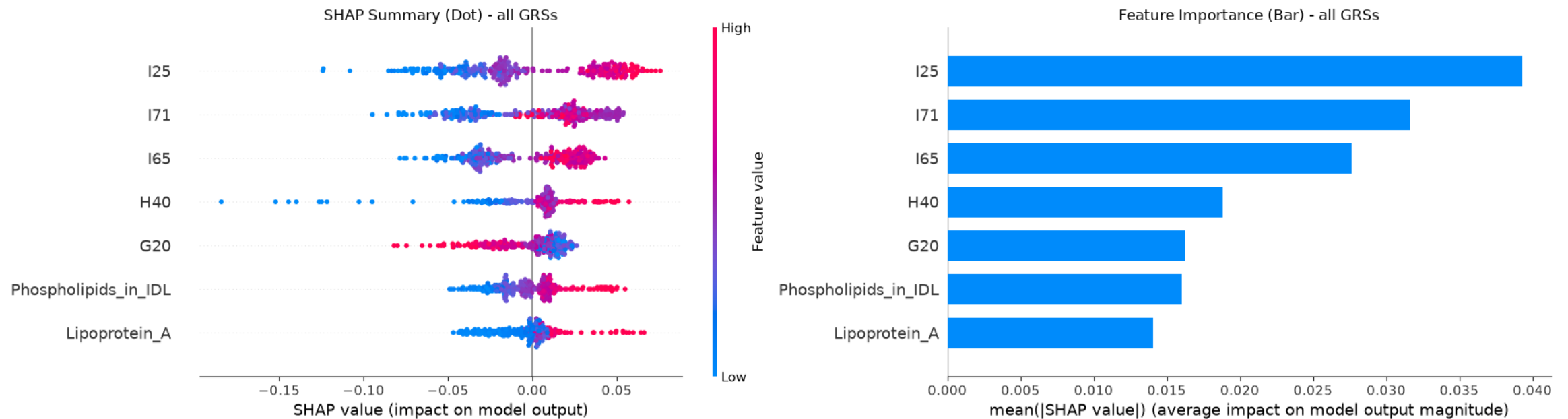
SHAP: the best model

- Best overall (Covs + all GRSs). Expect age and sex to dominate, since covariates alone already reach AUC around 0.82.

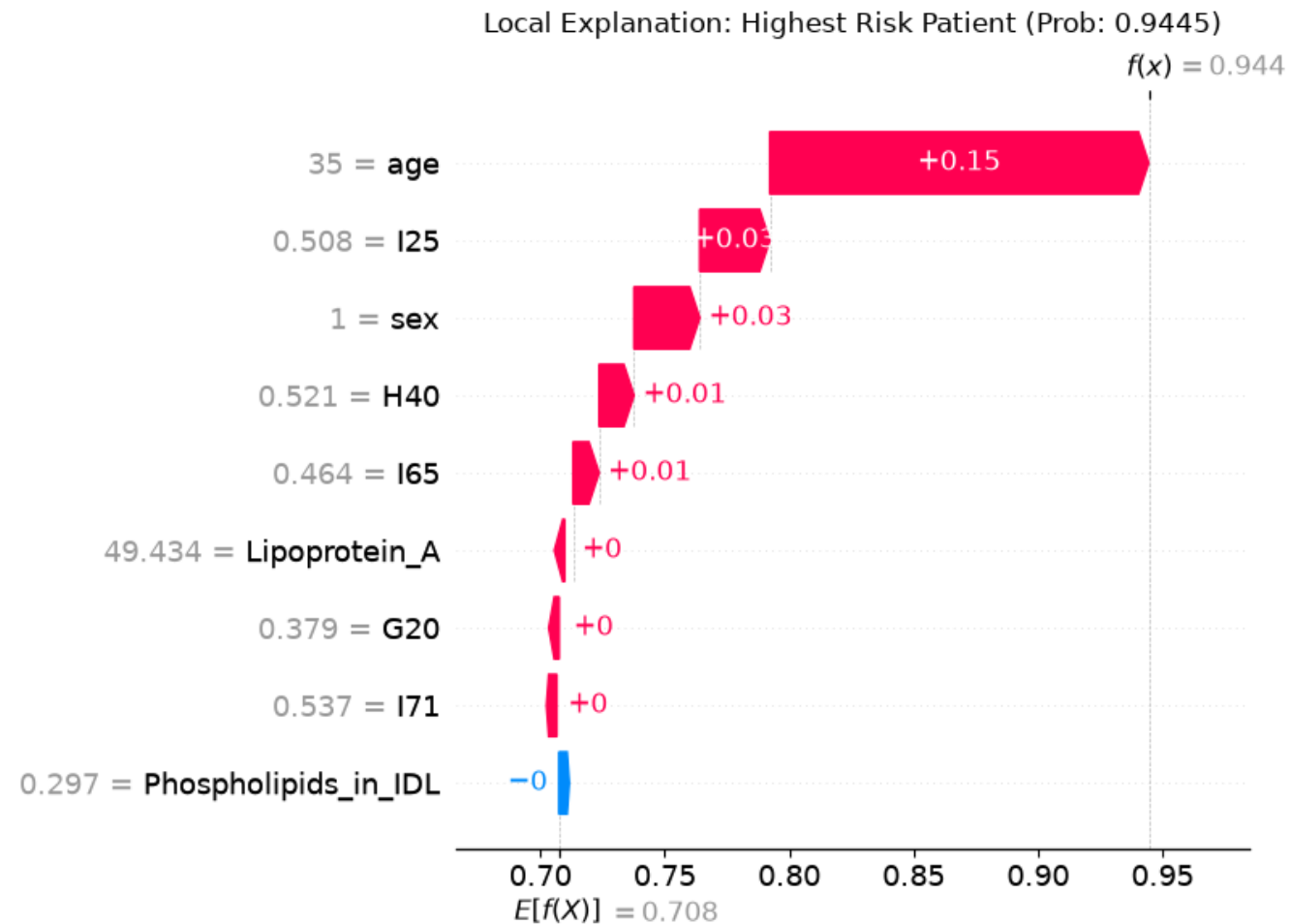


SHAP: the same panel without age and sex

- Pure GRS (all GRSs, no age or sex). This strips out the demographics so we can ask whether there is genetic signal beyond them.



One person: the highest-risk patient



A checklist for evaluating any genetic predictor

- Look at the labels first: prevalence, recruitment, missingness.
- Evaluate incrementally. ΔR^2 or ΔAUC over a baseline containing every confounder.
- Quantify uncertainty before ranking anything. Bootstrap CIs, repeated splits, or both.
- Explain the model, then check the explanation against biology.

Summary

- From PGS to a foundation model
- SNP harmonization
- Predicted against measured lipids
- Binary traits; ROC-AUC and precision-recall
- Correlations between the predicted traits
- Individuals in 2D. PCA and UMAP
- Downstream CAD prediction
- SHAP



Questions?